AI-ML-DS Learning Journey

Daily Learning Report — SQL Fundamentals

Date	20 August 2026
Learning Track	Artificial Intelligence / Machine Learning / Data Science
Current Module	SQL & Relational Databases
Focus	Database fundamentals, tables, queries, CREATE TABLE, and constraints

1. What I Learned Today

1. What is a Database? — Understood the purpose of a database as an organized system for storing, managing, and retrieving data efficiently.

2. SQL vs NoSQL — Learned the basic difference between relational SQL databases and non-relational NoSQL databases, including their different data models and typical use cases.

3. What is SQL? — Learned that SQL (Structured Query Language) is used to interact with relational databases for creating, reading, updating, and deleting data.

4. What is a Table? — Understood that a relational database stores data in tables made up of rows and columns.

5. Database Installation — Went through the initial database setup process for Windows and reviewed the installation process for Mac.

6. Creating the First Database — Practiced creating a database and understood the basic structure of a database environment.

7. Creating the First Table — Practiced creating a table with columns and data types.

8. Database Queries — Learned the role of SQL queries in communicating with and manipulating a database.

9. CREATE TABLE — Learned the basic syntax and purpose of the CREATE TABLE statement.

10. Constraints — Introduced to constraints that enforce rules on table data and help maintain data integrity.

2. Core SQL Concepts Covered

Concept	Understanding
Database	A structured collection of data that can be stored, managed, and queried.
SQL	A language used to interact with relational databases.
Table	A database object that organizes data into rows and columns.
Row	A single record in a table.
Column	A field/attribute describing a particular type of data.
Query	An SQL instruction used to retrieve or manipulate database information.

Concept	Understanding
CREATE TABLE	SQL command used to create a new table and define its columns.
Constraint	A rule applied to table data to maintain validity and integrity.

3. SQL Syntax Practiced

A basic table-creation statement follows this structure:

```
CREATE TABLE table_name (
    column_name data_type,
    column_name data_type
);
```

The important idea is that a table definition specifies the columns, their data types, and—when required—constraints that control what data can be stored.

4. Constraints — Initial Understanding

• PRIMARY KEY — Identifies each record uniquely.

• NOT NULL — Prevents a column from accepting NULL values.

• UNIQUE — Ensures values in a column are not duplicated.

• DEFAULT — Provides a default value when no value is supplied.

• CHECK — Applies a condition that values must satisfy.

• FOREIGN KEY — Defines a relationship between tables and helps maintain referential integrity.

5. Key Takeaways

• A relational database organizes information using tables.

• SQL is the primary language used to work with relational databases.

• Tables consist of rows and columns, and each column has a defined data type.

• SQL queries are the mechanism for interacting with database data and structures.

• Constraints are important because they prevent invalid or inconsistent data from entering a table.

• Today's learning establishes the foundation needed for SELECT queries, INSERT, UPDATE, DELETE, filtering, joins, aggregation, and database design.

6. Next Learning Focus

Continue with practical SQL operations: INSERT, SELECT, UPDATE, DELETE, WHERE conditions, ORDER BY, LIMIT, aggregate functions, GROUP BY, HAVING, and JOIN operations. The goal is to move from understanding database structure to confidently querying real datasets.

Learning status: SQL fundamentals completed for today's session. This report is prepared as a daily learning record for the AI-ML-DS journey and can be committed to GitHub as part of the learning history.